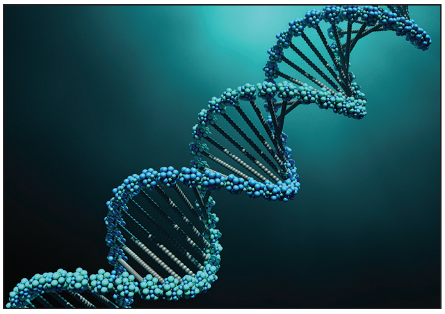
Examiner only

3. In the human genome, 99.9% of DNA is common in all individuals. The remaining 0.1% of the genome consists of DNA that is unique to individuals. During genetic profiling it is this unique 0.1% of the DNA that is separated and examined.

Image 3.1



Many online companies have websites that allow people to trace their ancestors and to build family trees. As part of this process they sell home DNA testing kits. Each kit contains a swab to remove cells from the inside of the user's cheek. The swab is placed in a container and sent to a laboratory where the DNA is extracted from the swab and processed for genetic profiling.

Image 3.2

(a) In **Image 3.2**, a swab is being used to remove cheek cells. The process is being carried out by another person and not by the person whose cheek cells are being collected. State why it is important for genetic profiling that the person using the swab is wearing gloves. [1]

....

....

(b) State what has to happen to DNA before it can be separated into bands during genetic profiling. [1]

....

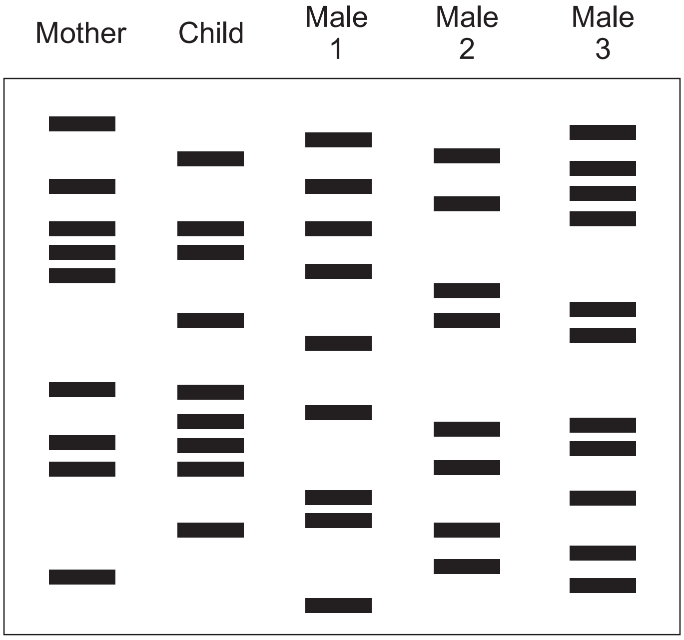
....

Examiner only

(c) Genetic profiles can be used in cases of disputed paternity. The genetic profile of both the mother and the child must be known. Any DNA band found in the genetic profile of the child that is not found in the genetic profile of the mother must be present in the genetic profile of the father for paternity to be confirmed.

Image 3.3 shows 5 genetic profiles, those of the mother and child, together with the profiles of three males who are disputing paternity of the child.

Image 3.3



(i) Using **Image 3.3**, state which male is the father of the child. [1]

....

(ii) Support your answer by **circling three** DNA bands, on the profile of the father in **Image 3.3**, that proves your answer. [2]

(d) Apart from disputed paternity and tracing ancestors, give **two** other uses of genetic profiling. [2]

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